

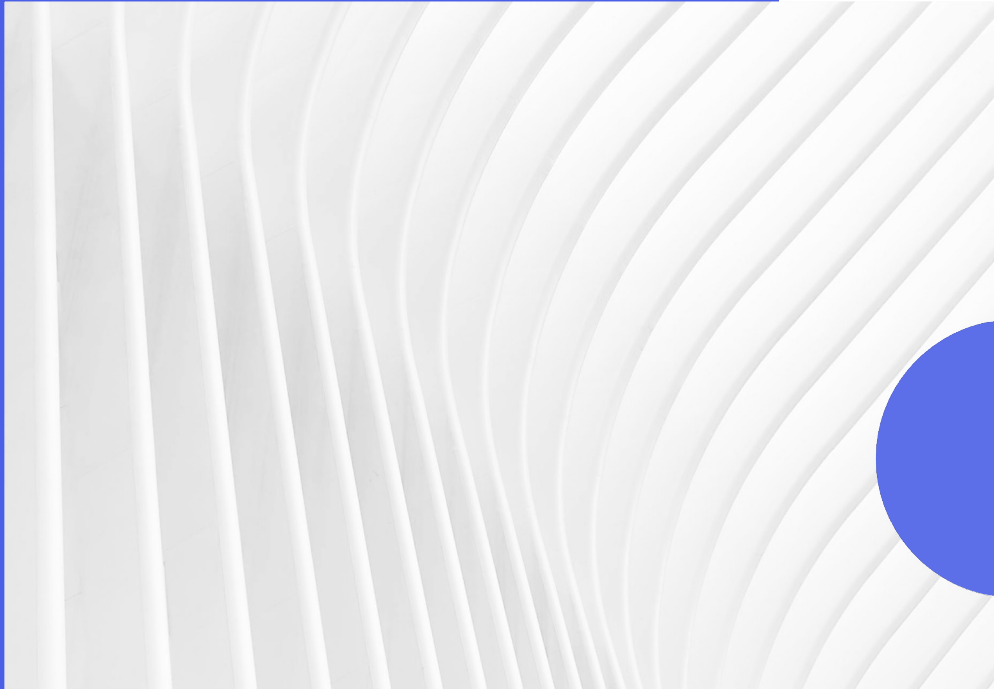
BAYESIAN INFERENCE

**A MATH-FREE*
INTRODUCTION**

JAKE BARLOW



INTRO



- Philosophy of reason
- Comparison of approaches
- Who (and what) is Bayes(ianism)
- Interpretation
- Implementation



STATISTICAL REASONING

DOCTRINES OF CHANCE

Two competing philosophies of probability:

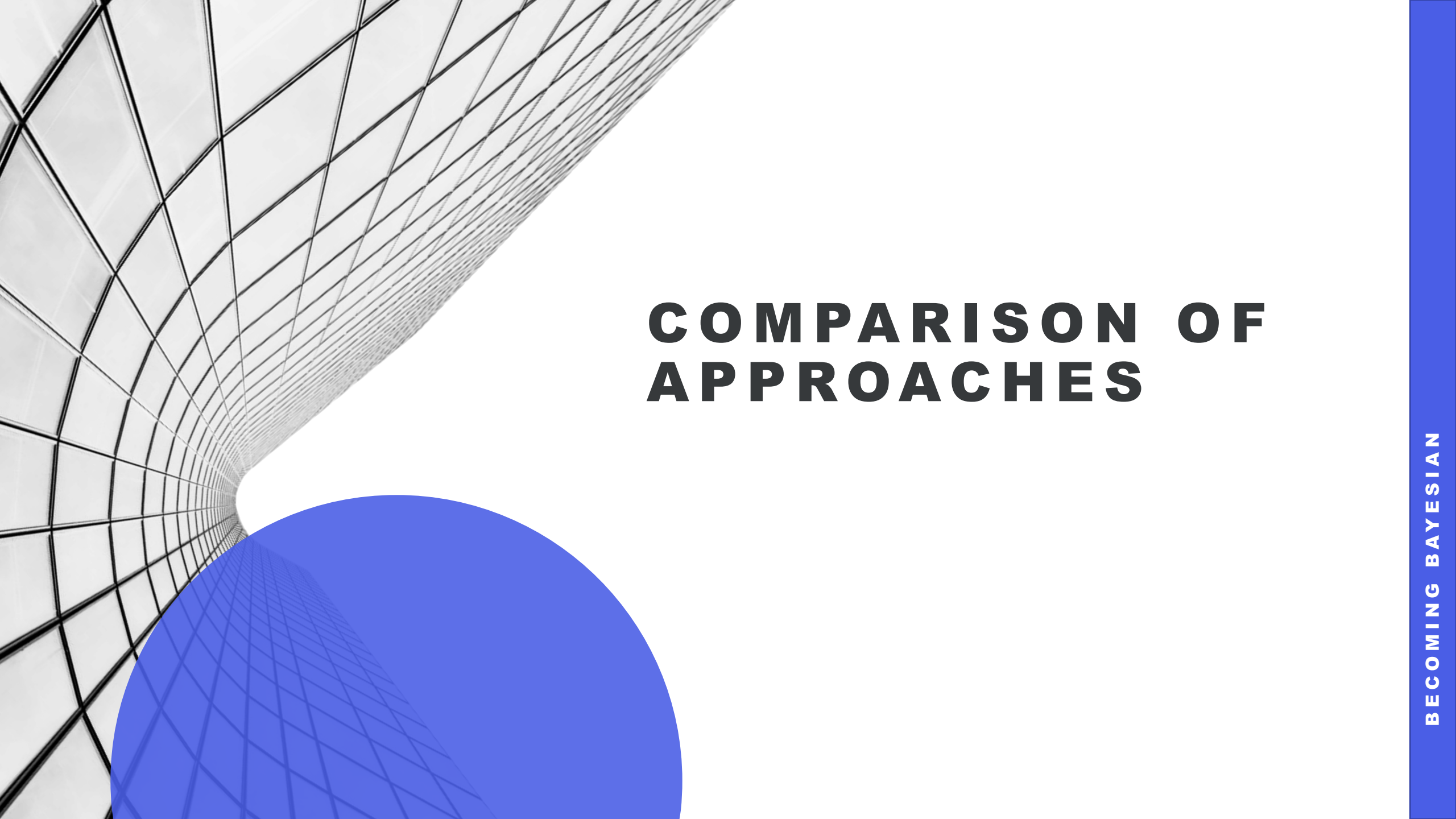
1. Objective/frequentist

An event has a specific, but unknown, chance of occurring.

2. Subjective/Bayesian

An event has an unknown chance of occurring...

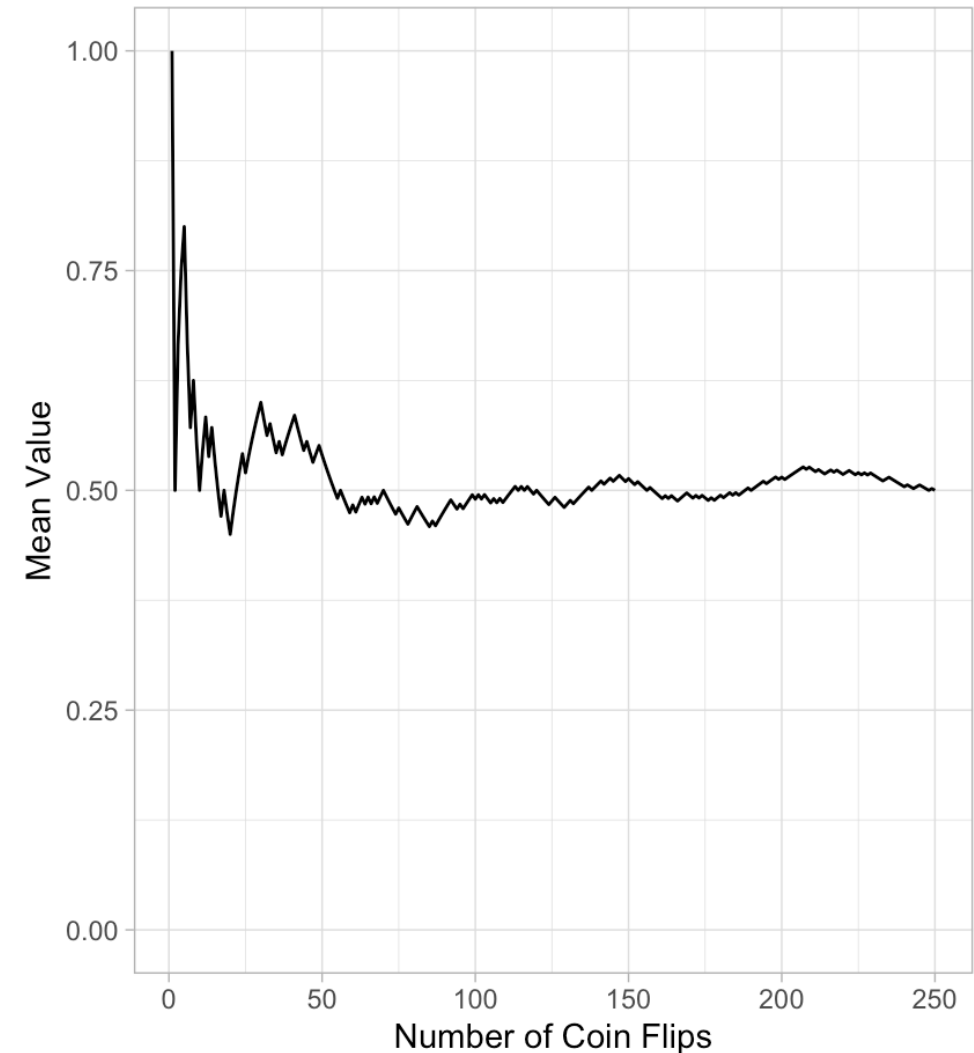
...but some chances are more likely than others.



COMPARISON OF APPROACHES

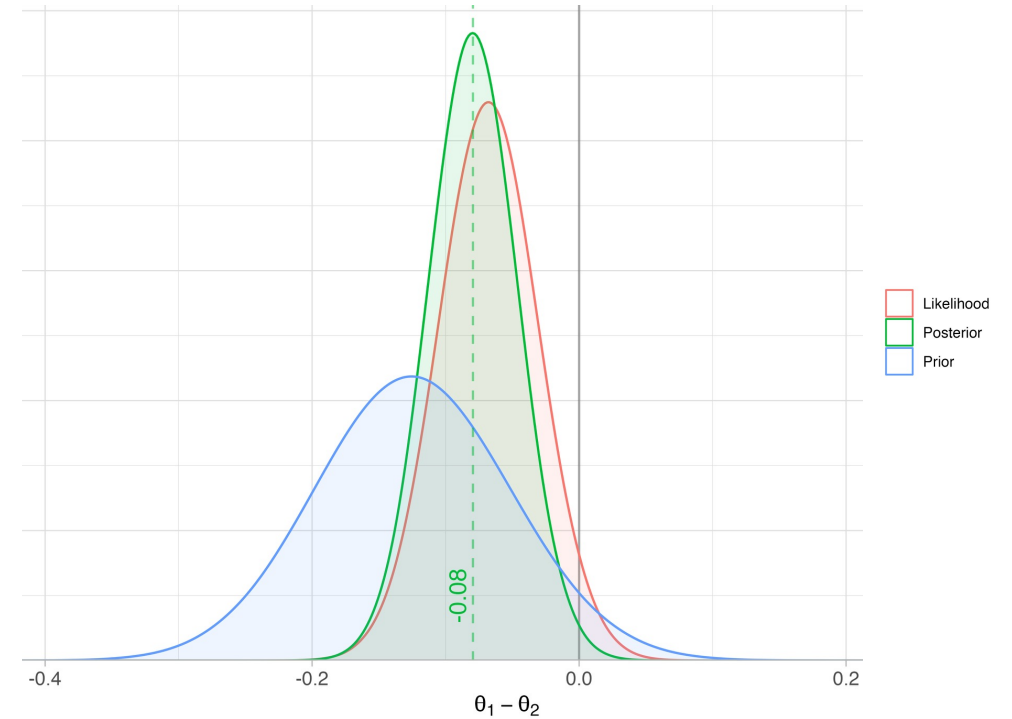
FREQUENTISM

- Belief: Probability of an event is a relative frequency over time
- e.g. Probability of heads on a fair coin is 0.5



BAYESIANISM

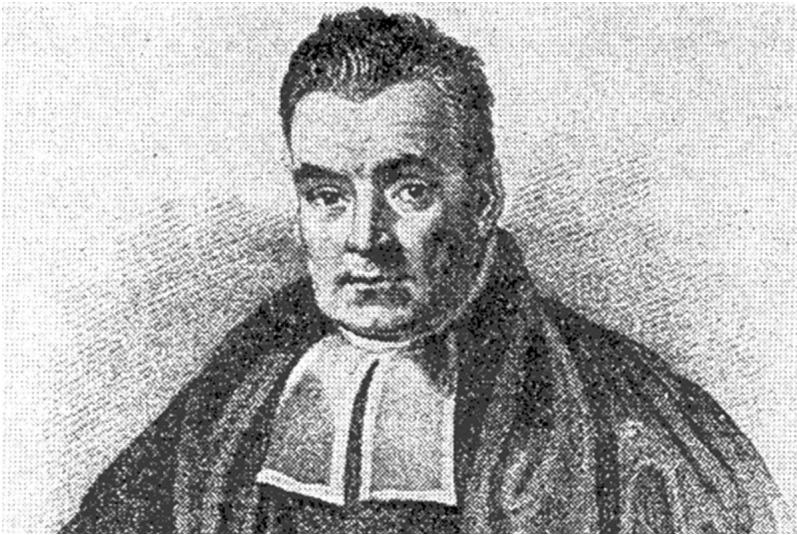
- Belief: Probability of an event is uncertain, but degree of uncertainty reduces with more information
- E.g. The size of a mortality reduction given some treatment





BAYESIANISM

YOU SIMPLY MUST MEET THOMAS



- Rev. Thomas Bayes (1701-1761)
- Presbyterian minister
- Developed Bayes Theorem

$$P(\text{Hypothesis}|\text{Data}) = P(\text{Hypothesis}) \times \frac{P(\text{Data}|\text{Hypothesis})}{P(\text{Data})}$$

WHY?

Frequentist:

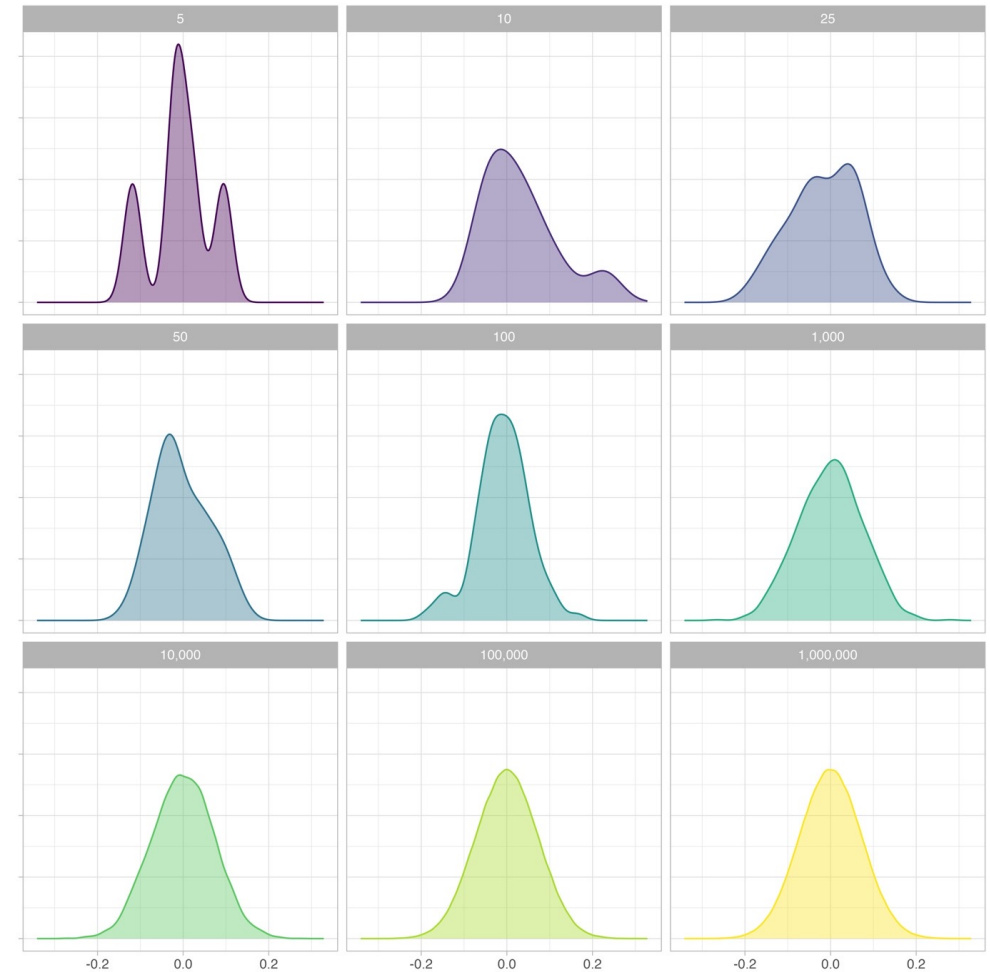
"How likely is this (or more extreme) data, if my hypothesis is incorrect?"

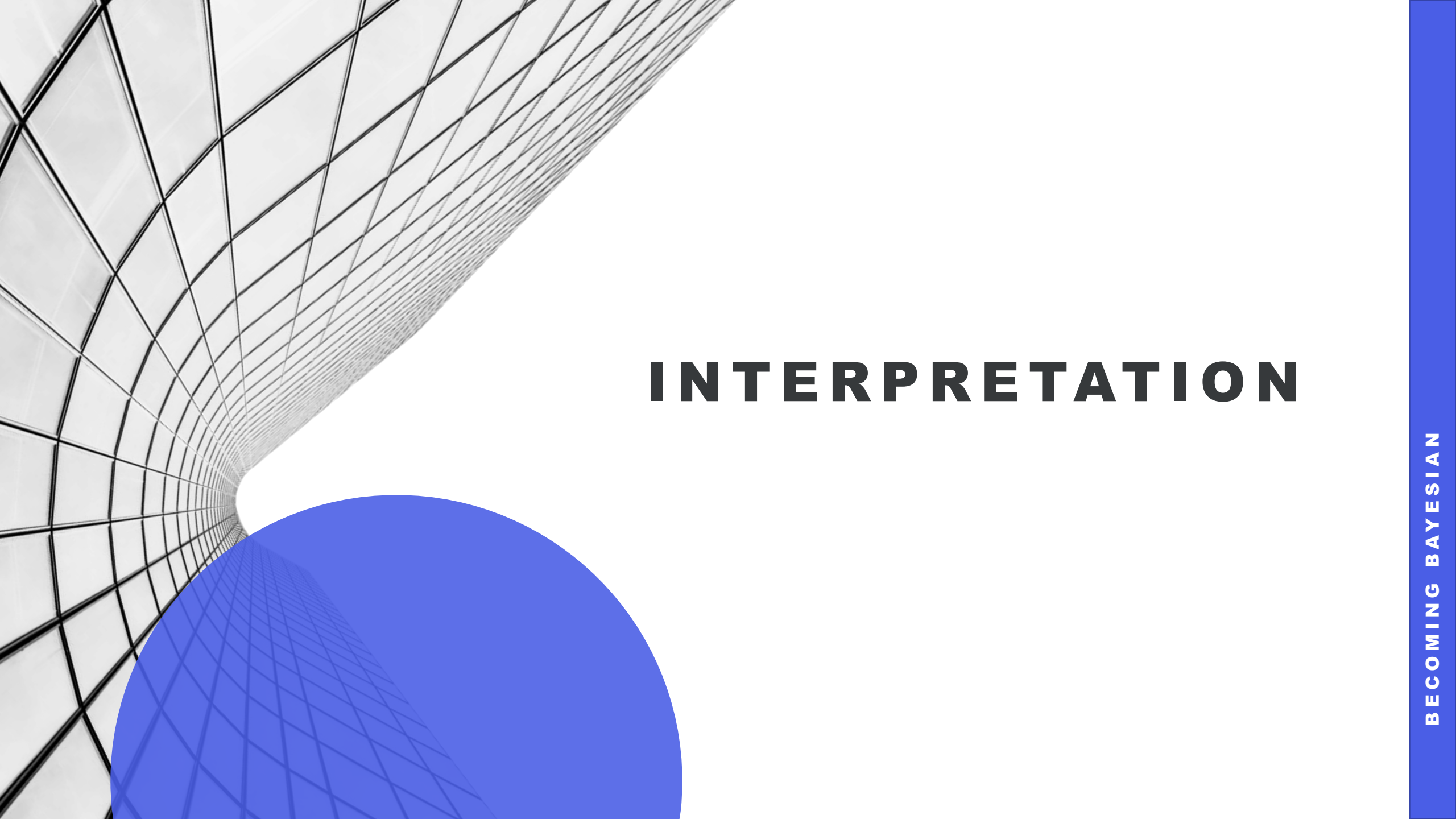
Bayesian:

"How likely is my hypothesis, given this data."

WHY NOW?

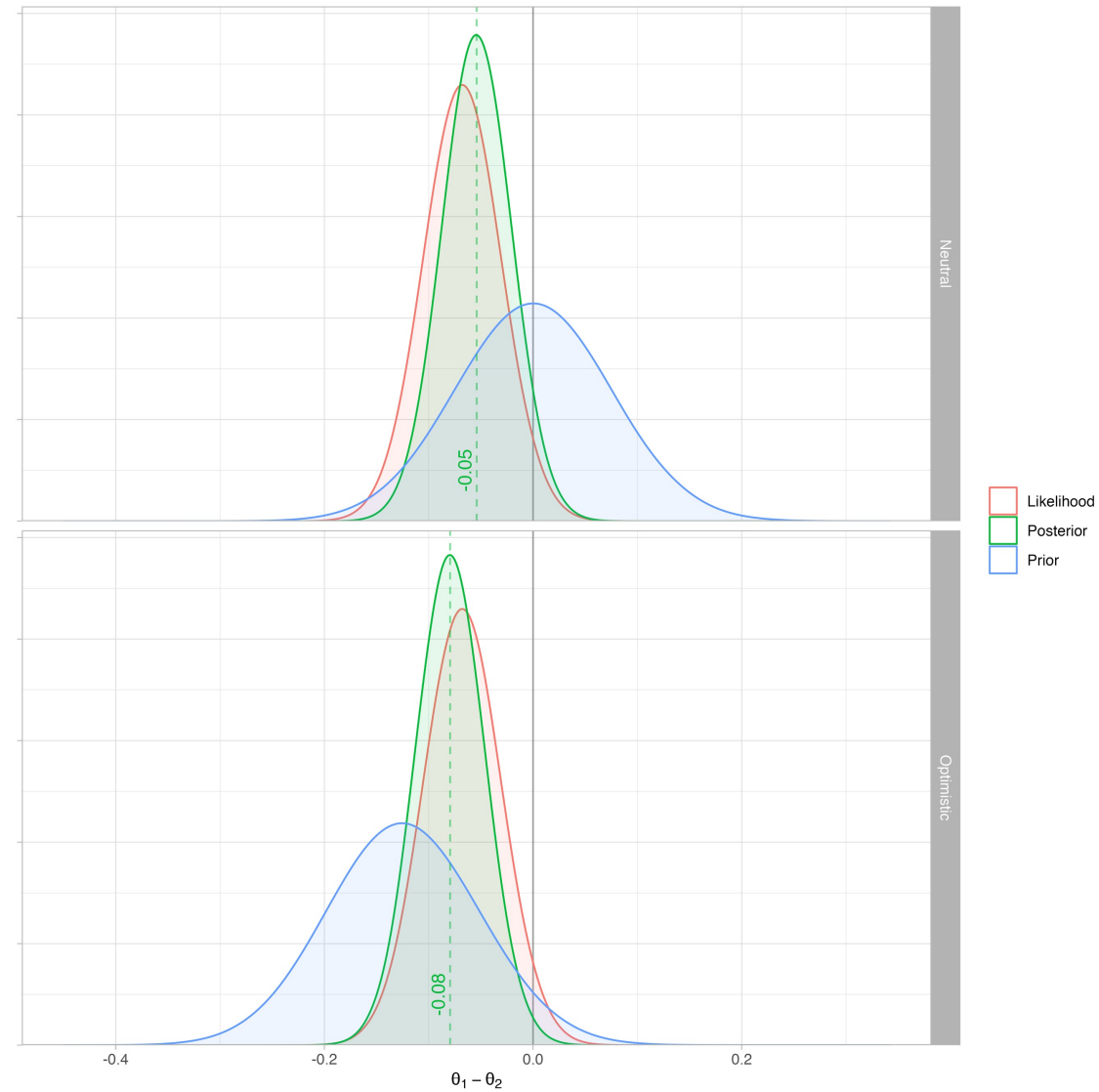
- Philosophical
- Sociological
- Technological





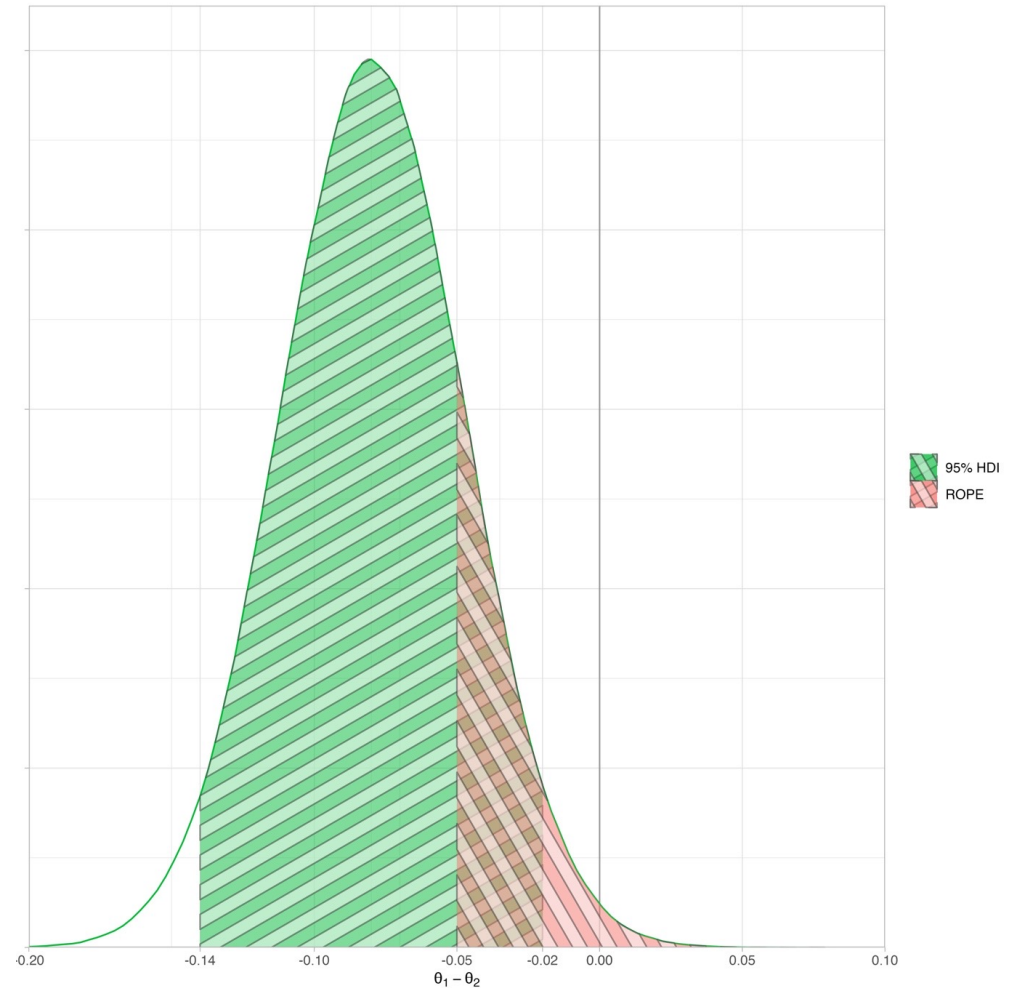
INTERPRETATION

BAYESIAN MODELS



INTERVALS

- Highest Density Interval (HDI)
- Region of Practical Equivalence (ROPE)





THANK YOU

Jake Barlow